CrowdWisdomTrading: Quantitative Prediction Report

Multi-Horizon Evaluation across 70% Dev & 30% Holdout | Generated 2026-10-19 17:06 UTC

1. Research Methodology & Leakage Prevention

All horizons predict return from confirmed close(t-1) to future close(t+h). Training features use information strictly up to close(t-1). Evaluated using 70% Development (purged walk-forward CV with 5-bar embargo) and 30% Frozen Chronological Holdout.

2. Development Period Walk-Forward Validation Metrics (70% Window)

Horizon	Model	MAE ($)	RMSE ($)	Dir Acc (%)	Base MAE ($)	DM p-val
n+0	Persistence	$185.2	$292.7	49.2%	$185.2	1.000
n+0	Ridge	$185.5	$293.4	51.3%	$185.2	1.000
n+0	Random Forest	$185.7	$293.7	51.3%	$185.2	1.000
n+0	Hist Gradient	$186.3	$294.2	51.7%	$185.2	1.000
n+1	Persistence	$263.7	$407.3	49.1%	$263.7	1.000
n+1	Ridge	$264.6	$408.3	52.5%	$263.7	1.000
n+1	Random Forest	$265.1	$410.2	51.7%	$263.7	1.000
n+1	Hist Gradient	$265.3	$410.2	52.1%	$263.7	1.000
n+2	Persistence	$323.0	$492.1	49.2%	$323.0	1.000
n+2	Random Forest	$323.8	$495.4	52.8%	$323.0	1.000
n+2	Ridge	$324.8	$494.2	51.2%	$323.0	1.000
n+2	Hist Gradient	$325.1	$496.3	51.8%	$323.0	1.000
n+3	Persistence	$370.5	$558.4	50.0%	$370.5	1.000
n+3	Random Forest	$374.5	$564.3	50.1%	$370.5	1.000
n+3	Ridge	$377.1	$564.8	50.6%	$370.5	1.000
n+3	Hist Gradient	$377.6	$567.7	50.5%	$370.5	1.000
n+4	Persistence	$414.8	$616.4	49.9%	$414.8	1.000
n+4	Random Forest	$422.3	$625.9	50.8%	$414.8	1.000

3. Frozen Holdout Validation Metrics (30% Quarantined Window)

Horizon	Model	MAE ($)	RMSE ($)	Dir Acc (%)	Base MAE ($)	DM p-val
n+0	Persistence	$191.2	$291.6	50.9%	$191.2	0.998
n+0	Random Forest	$191.4	$291.6	50.3%	$191.2	0.998
n+0	Hist Gradient	$191.8	$292.6	49.5%	$191.2	0.998
n+0	Ridge	$192.7	$293.2	50.9%	$191.2	0.998
n+1	Persistence	$270.3	$413.5	52.4%	$270.3	0.998
n+1	Random Forest	$271.4	$414.4	50.4%	$270.3	0.998
n+1	Hist Gradient	$272.9	$415.8	49.1%	$270.3	0.998
n+1	Ridge	$274.2	$416.9	50.0%	$270.3	0.998
n+2	Random Forest	$334.0	$502.5	52.9%	$335.0	0.998

n+2	Persistence	$335.0	$501.5	51.1%	$335.0	0.998
n+2	Hist Gradient	$336.8	$504.8	53.1%	$335.0	0.998
n+2	Ridge	$339.6	$507.7	51.2%	$335.0	0.998
n+3	Random Forest	$386.1	$578.2	55.3%	$387.9	0.998
n+3	Persistence	$387.9	$577.0	52.1%	$387.9	0.998
n+3	Hist Gradient	$388.3	$580.6	55.1%	$387.9	0.998
n+3	Ridge	$392.0	$583.9	51.7%	$387.9	0.998
n+4	Random Forest	$436.6	$643.5	53.3%	$437.0	0.998
n+4	Persistence	$437.0	$639.6	52.3%	$437.0	0.998

4. Quantitative Trading Strategy Performance (10 bps Friction)

Model	Horizon	Split	Cum Ret	Exposure	Sharpe	Sortino	Max DD	Trades
No Trade Bas	n+1	Dev_Walk	+0.0%	0.0%	0.00	0.00	0.0%	0
Ridge	n+1	Dev_Walk	-7.0%	4.3%	-3.20	-4.67	-7.9%	110
Random Fores	n+1	Dev_Walk	-6.0%	4.0%	-2.96	-4.29	-10.0%	80
Hist Gradien	n+1	Dev_Walk	-5.1%	3.3%	-2.38	-3.73	-9.3%	72
No Trade Bas	n+1	Frozen_H	+0.0%	0.0%	0.00	0.00	0.0%	0
Ridge	n+1	Frozen_H	-8.3%	6.3%	-6.01	-7.24	-8.4%	64
Random Fores	n+1	Frozen_H	-2.2%	5.7%	-2.16	-3.15	-3.3%	38
Hist Gradien	n+1	Frozen_H	-1.9%	6.1%	-1.97	-2.78	-2.6%	30